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The Neurocritical Care Examination and Workup

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ABSTRACT

OBJECTIVE: This article provides an overview of the evaluation of patients in neurocritical care settings and a structured approach to recognizing and localizing acute neurologic emergencies, performing a focused examination, and pursuing workup to identify critical findings requiring urgent management.

LATEST DEVELOPMENTS: After identifying and stabilizing imminent threats to survival, including respiratory and hemodynamic compromise, the initial differential diagnosis for patients in neurocritical care is built on a focused history and clinical examination, always keeping in mind critical “must-not-miss” pathologies. A key priority is to identify processes warranting time-sensitive therapeutic interventions, including signs of elevated intracranial pressure and herniation, acute neurovascular emergencies, clinical or subclinical seizures, infections of the central nervous system, spinal cord compression, and acute neuromuscular respiratory failure. Prompt neuroimaging to identify structural abnormalities should be obtained, complemented by laboratory findings to assess for underlying systemic causes. The indication for EEG and lumbar puncture should be considered early based on clinical suspicion.

ESSENTIAL POINTS: In neurocritical care, the initial evaluation is often fast paced, requiring assessment and management to happen in parallel. History, clinical examination, and workup should be obtained while considering therapeutic implications and the need for lifesaving interventions.

INTRODUCTION

Acute neurologic emergencies are common and potentially devastating, with high mortality and risk of substantial long-term disability.¹⁻³ Prompt recognition is crucial because there are critical time windows for therapeutic interventions to improve neurologic function⁴⁻⁸ or prevent secondary neurologic injury.⁹ There is potential for rapid, life-threatening deterioration, often manifesting with changes in the neurologic examination, which herald the need for emergent interventions.

A recent meta-analysis demonstrated improved survival and neurologic outcomes in patients with acute brain injury who received subspecialized neurocritical care services, highlighting the importance of neurologic expertise and subspecialty care for patients with acute neurologic emergencies.¹⁰

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However, geographic disparities in access to neurocritical care¹¹ and variations in the initial care and workup of patients with neurologic emergencies have been described.^{12,13}

This article provides a structured diagnostic approach to the initial evaluation of patients in neurocritical care settings, including the recognition of critical pathology, relevant clinical examination findings, and the use of diagnostic tests to guide critical and time-sensitive treatment decisions.

PRIORITIES FOR THE INITIAL EVALUATION

When evaluating patients with acute neurologic deficits, the key is to identify the imminent threats to the patient, recognize and localize the problem, and, foremost, consider the need for emergent interventions.

The initial evaluation should begin with an assessment of imminent threats to survival, including respiratory and hemodynamic compromise. This should include the evaluation of airway patency and adequacy of ventilation, oxygenation, circulation, and perfusion. Although the phrase “if the GCS score is less than 8, then intubate” is commonly repeated, tracheal intubation for depressed consciousness alone is not recommended and may be harmful.¹⁴⁻¹⁷ Objective signs of airway compromise include visible airway obstruction, pooling secretions, or upper airway noises such as snoring, stridor (noisy breathing at or below the larynx), or stertor (noisy breathing above the larynx). Using the gag reflex to determine the need for tracheal intubation is controversial, although inadequate cough is a common concern in neurocritically ill patients. Adequacy of ventilation may be assessed by direct observation, auscultation, waveform capnography, and blood gas analysis. For patients with suspected neuromuscular weakness, maximum inspiratory pressure (also known as *negative inspiratory force*, normally greater [more negative] than -60 cm H₂O), maximum expiratory pressure (normally greater than 100 cm H₂O), and vital capacity (range 60 mL/kg to 70 mL/kg) measurement as surrogates for adequate ventilation may be considered. Oxygenation can be assessed via pulse oximetry or blood gas analysis, ideally without supplemental oxygen.

Hemodynamic compromise may be evaluated via blood pressure and heart rate, although it is important to note that these can be normal in early compensated shock. Mental status, a helpful indicator of adequate cerebral perfusion, may be confounded by brain injury. Extremity temperature, mottling, capillary refill, and urine output are other markers of adequate perfusion. As the primary survey is completed, interventions to treat any derangements, such as tracheal intubation if airway compromise is found, should be performed in tandem with assessment. Key considerations related to the neurocritical patient include, ideally, obtaining a baseline, focused neurologic examination before intubation and sedation or considering choices of sedation and neuromuscular blocking agents that can preserve the examination or be easily reversed. As part of the primary survey, neurologists should consider rapidly reversible confounders, such as hypoglycemia, which may mimic a variety of neurologic conditions.¹⁸

The initial differential diagnosis is built on a focused history and clinical examination. Presenting symptoms, time course, pace of progression, and evolution of symptoms provide important clues about the nature of the process (“What is the problem?”), whereas the description of symptoms and the constellation of neurologic examination findings determine the localization of

KEY POINT

● The initial evaluation of patients in the neurocritical care setting begins with the assessment and stabilization of imminent threats to survival and includes evaluation of airway patency, adequacy of ventilation and oxygenation, and adequate circulation and perfusion.

the underlying process (“Where is the problem?”) on the neuraxis. Establishing the *what* and *where* early on will guide additional workup, including neuroimaging, laboratory studies, EEG, and lumbar puncture (LP).

In reality, however, the initial evaluation is often a dynamic process. The history may be incomplete or false, key information may need verification repeatedly, and pertinent updates may shift the direction of the workup. The clinical examination may suddenly change, requiring rapid readjustments to the treatment plan. At times, critical results of neuroimaging or laboratory tests precede the anamnesis and will have to be addressed while the information-gathering process is ongoing. Ultimately, the assessment typically must happen in parallel with management, prioritizing treatment of imminent life threats. As a key modality to identify structural lesions, neuroimaging should be obtained promptly in patients with suspected acute brain injury to determine the need for urgent neurosurgical interventions, especially in the face of rapid clinical deterioration. Meanwhile, identifying the risk of deterioration and ensuring appropriate monitoring while the evaluation is ongoing are crucial when caring for patients with neurologic emergencies.

Based on the initial clinical impression and consideration of treatment priorities, clinicians must determine whether to formulate a concise assessment (the problem is relatively clear and time sensitive) or pursue a more comprehensive workup (the problem is not clear and less time sensitive). To establish the most appropriate approach, it is important to recognize key characteristics and patterns of neurologic emergencies warranting time sensitive treatments, such as acute ischemic stroke or infections of the central nervous system (CNS).

While homing in on the most likely diagnosis, the list of differential diagnoses should always include all potentially life-threatening conditions. These conditions may be quickly excluded based on clinical presentation or diagnostic test results but should always be considered. **TABLE 1-1** provides an overview of “must-not-miss” neurocritical care emergencies and the most concerning manifestations of these conditions.

ELICITING THE STORY

Taking the history is a critical step in forming a differential diagnosis and guiding the initial workup, but it may not be an easy task. In critically ill patients with high acuity, the history is frequently gathered in parallel to the workup amid a hectic pace. Initial snippets of information require verification and clarification via subsequent inquiries for additional details. Many patients with acute neurologic emergencies are not able to provide a history, and clinicians must obtain collateral information from family, bystanders, emergency medical services personnel, other clinicians, and recent chart documentation. Family members and layperson witnesses may have different perceptions and understandings of medical terms and clinical symptoms. Additionally, the stressful experience of having to describe a medical history while worrying about a family member is often underappreciated by clinicians. To gather accurate and relevant information, clinicians must recognize the pertinent facts, ask the right questions and steer the interview in the right direction, interpret and self-correct what is presented, and synthesize all important details. These skills require experience and knowledge of disease. When forming an assessment, a key challenge is to focus on the most likely diagnosis while considering the most

life-threatening alternative diagnoses and simultaneously keeping an open mind and acknowledging the potential for biases, most notably anchoring bias, and diagnostic errors.¹⁹

Signs or symptoms that should always raise concerns for an acute neurologic emergency include (1) rapid decline in level of consciousness; (2) new confusion, behavioral changes, or agitation; (3) new motor weakness or sensory deficit; (4) new language or speech disturbance; (5) new inability to walk or balance difficulties; (6) new visual symptoms; (7) new involuntary movements or posturing; and (8) new-onset severe headaches. Establishing critical time windows early on, such as the time last seen well in acute ischemic stroke, is important to determine eligibility for interventions and facilitate prompt treatment. Another crucial aspect of obtaining a complete history is to exclude concerning pathology. **TABLE 1-2** reviews subtle red flags for critical diagnoses, the importance of which is illustrated in **CASE 1-1**.

To identify the underlying disease process, establishing the onset and evolution of symptoms over time is essential. Neurologic disease entities can be categorized based on symptom onset into hyperacute (seconds to minute), acute (hours to days), subacute (weeks to months), and chronic (years). **FIGURE 1-2** delineates the time course of neurologic diseases with emphasis on neurologic emergencies. Of note, some subacute disease processes can present acutely (eg, a slowly growing mass may evolve over months and suddenly manifest with an intracranial hemorrhage [ICH] or a seizure). The evolution of the symptoms can be stable, fluctuating, progressive, or episodic, providing additional clues about the diagnosis. Finally, the differential diagnosis can be further narrowed by inquiring about pertinent positive and negative characteristics of suspected etiologies.

As is relevant to the assessment of all neurologic conditions, it is essential to think about localization along the neuraxis. Is the lesion within the CNS, and if so, does it localize to the brain, brainstem, cerebellum, or spinal cord? Or is it a process within the peripheral nervous system (PNS) at the level of the spinal nerve root, dorsal root ganglia, peripheral nerve, neuromuscular junction, or muscle? Localization can then be teased out on a more granular level (structures and tracts) by exploring the nature of the presenting symptom and constellation of other neurologic symptoms, coupled with findings on the neurologic examination.

Getting to know the patient's background and prior history provides additional context that can guide diagnosis and treatment. Pertinent comorbidities in patients in neurocritical care include cardiovascular risk factors, history of prior thromboembolic events, prior history of stroke, prior history of epilepsy or seizures, recent or remote neurosurgical procedures (including ventriculoperitoneal shunt placement), psychiatric history, history of autoimmune disease, and substance use or withdrawal. Medications that are often relevant in this population include anticoagulation, antiplatelet agents, antiseizure medications, immunosuppressive agents, antipsychotics, sedative-hypnotics, and analgesics. In hospitalized patients with new neurologic concerns, the presence and timing of sedation and neuromuscular blockade must be established while considering renal or hepatic clearance of these medications. Neurotoxicity and neurologic side effects of prescribed medications should also be considered.

Finally, another important purpose of the history and information gathering in patients in neurocritical care is to guide life-or-death,

KEY POINTS

- Information about the time course, the evolution of the presenting symptoms, and a focused clinical examination in the neurocritical care setting are key to identifying (what) and localizing (where) the underlying problem.
- While the neurocritical care evaluation is ongoing, the list of differential diagnoses should always include a list of life-threatening, "must-not-miss" conditions, considering implications for management.
- Assessment of patients in neurocritical care often must occur at a fast pace and in parallel with management, establishing critical time windows for time-sensitive interventions early on. The clinical history must be verified and updated and should consider the patient's overarching goals of care and core values to guide emergent decision making.

time-sensitive treatment decisions. Patients with acute neurologic emergencies often require urgent interventions, including neurosurgical procedures (eg, decompressive craniectomy to treat elevated intracranial pressure (ICP), diversion of CSF with an extraventricular drain, endovascular procedures), and tracheal intubation. Gaining insights about the patient's overarching goals of care, including personhood, core values, and prior wishes, and their baseline functional status before admission is crucial to determining whether to consider and how to discuss invasive life-sustaining treatments.²⁰

PINPOINTING PATTERNS IN THE CLINICAL EXAMINATION

The goal of the neurologic examination is to guide localization by discerning patterns corresponding to various structures and levels of the neuraxis. Findings on the examination can often confirm or disprove hypotheses formed based on

TABLE 1-1 Neurologic Emergencies

Localization	Underlying condition	Common acute threats and critical concerns
Brain ^a	Traumatic brain injury	Mass lesion or intracranial hemorrhage (ICH), diffuse cerebral edema leading to elevated intracranial pressure (ICP) and herniation; blunt cerebrovascular injury causing ischemic stroke; facial trauma, airway compromise; polytrauma (pneumothorax, perforation, penetrating injuries); long bone fractures leading to cerebral fat emboli syndrome; hydrocephalus; paroxysmal sympathetic hyperactivity, autonomic instability
	Acute ischemic stroke	Large vessel occlusion causing worsening cerebral ischemia; insufficiently perfused penumbra leading to worsening cerebral ischemia; malignant cerebral edema causing herniation (middle cerebral artery, cerebellum), hydrocephalus (cerebellum); cardiovascular comorbidities (heart failure, cardiac arrhythmias, intracardiac thrombus)
	Spontaneous intracranial hemorrhage	ICH expansion; mass effect leading to elevated ICP and herniation; intraventricular extension of ICH leading to hydrocephalus
	Subarachnoid hemorrhage	Aneurysmal rerupture; cardiac arrhythmias, hemodynamic instability; hydrocephalus; postprocedure (coil or clip) complications; neurogenic cardiomyopathy, cardiogenic shock; cerebral vasospasm; delayed cerebral ischemia
	Post-cardiac arrest syndrome and hypoxemic ischemic encephalopathy	Recurrent cardiac arrhythmias and cardiac arrest; underlying arrest causes (eg, myocardial infarctions, pulmonary embolism, pneumothorax, cardiac tamponade); hemodynamic instability; acute respiratory distress syndrome (ARDS); complications of therapeutic hypothermia; seizures and myoclonus; cerebral edema causing elevated ICP and herniation
	Venous sinus thrombosis	Venous congestion leading to worsening edema, ICH, and herniation; underlying systemic causes of hypercoagulable state
	Status epilepticus	Permanent neurologic injury; ictal cardiac arrhythmias; side effects of antiepileptic medications and sedatives
	Posterior reversible encephalopathy syndrome (PRES)	Worsening and permanent neurologic deficit; cerebral ischemia and ICH; complications related to underlying cause (eclampsia, fetal distress, transplant rejection)

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the history and, at times, generate new considerations. In acute neurologic emergencies, a focused examination is foremost aimed at detecting signs of concerning processes that warrant intervention, including signs of elevated ICP and herniation, acute neurovascular emergencies, clinical or subclinical seizures, infections of the CNS, spinal cord compression, and acute neuromuscular respiratory failure.

There are several distinctive considerations for patients in neurocritical care. First, the neurologic examination is often limited by the patient's inability to cooperate because of an impaired level of consciousness. Brainstem reflexes, eye position, and motor responses to noxious stimuli are the most critical components of the examination in this scenario. At times, noninvasive and invasive monitoring techniques can be used to complement or supplement the examination.^{21,22} Second, the neurologic examination may rapidly change in patients with acute neurologic emergencies, and the potential for deterioration

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Localization	Underlying condition	Common acute threats and critical concerns
Brain^a	Central nervous system (CNS) infections	Mass effect or edema causing elevated ICP and herniation; vasculitis, ischemic and hemorrhagic strokes; hydrocephalus; complications related to systemic infection
	CNS malignancies	Mass effect or edema causing elevated ICP and herniation; cerebral ischemia and ICH; hydrocephalus; coagulopathy; complications related to systemic malignancy
	Autoimmune CNS disorders	Neurologic deterioration; vasculitis, ischemic and hemorrhagic strokes; side effects of immunosuppressive medications, infection
	Toxidromes	ECG changes, cardiac arrhythmias; seizures, withdrawal; serotonin syndrome, neuroleptic malignant syndrome, malignant hyperthermia
Spinal cord	Traumatic spinal cord injury	Cord compression (external [bone], internal [syrinx]); hemodynamic instability (cervicothoracic cord), bradycardia, asystole; hypoventilation, inadequate secretion management, diaphragmatic weakness leading to acute respiratory failure
	Spinal cord infections, malignancies, autoimmune or inflammatory disease	Hemodynamic instability (cervicothoracic cord); bradycardia, asystole
Peripheral nerve	Acute inflammatory demyelinating polyradiculoneuropathy (AIDP), acute motor axonal neuropathy (AMAN)	Hypercarbic respiratory failure, respiratory acidosis; autonomic instability, cardiac arrhythmias
Neuromuscular junction	Myasthenia gravis or botulism	Hypercarbic respiratory failure, acidosis, myasthenic crisis, cholinergic crisis

^a Common concerns in all acute brain injuries are acute respiratory failure due to insufficient airway protection and seizures in patients with cortical pathology.

must be identified to determine the optimal frequency of evaluations based on initial examination and imaging findings. Third, when performing a neurologic examination, multiple confounders must be considered in critically ill patients. The examination should ideally be performed off sedation and in the absence of neuromuscular blockade, although at times this may not be feasible (eg, when sedation is used for control of elevated ICP or refractory status epilepticus). Type and timing of last sedative medications, presence of renal and hepatic failure, and clearance of sedative medications must be considered. If neuromuscular blockade has been administered within 5 half-lives (or longer in the presence of hepatic or renal dysfunction), train-of-four monitoring with a peripheral nerve

TABLE 1-2 Subtle Red Flag Signs and Symptoms Indicating Life-Threatening Conditions

Diagnosis	Signs and symptoms	Potential threats
Basilar artery occlusion	Dizziness and vertigo; diplopia, blurry vision; gaze palsies, (vertical) skew deviation; central nystagmus; behavioral changes; crossed findings: ipsilateral cranial nerve, contralateral motor symptoms	Brainstem infarction, locked-in syndrome, death; dysregulation of breathing control; aspiration, acute respiratory failure; autonomic dysregulation
Cerebellar infarct	Dizziness and vertigo; nausea and vomiting; central nystagmus; dysmetria, intention tremor; hiccups; Wallenberg syndrome; wide-based gait, truncal ataxia; falling to one side	Brainstem injury, herniation; fourth ventricular obstruction leading to hydrocephalus
Meningitis or encephalitis	Headaches; altered mental status; fever; meningismus, photophobia; malaise; infectious prodromes (eg, cough, diarrhea)	Long-term neurologic sequelae; seizures and status epilepticus; intracranial pressure (ICP) crises and herniation; temporal lobe hemorrhage (herpes simplex virus encephalitis); cerebral vasospasm causing ischemic infarcts; systemic infection, sepsis
Intracranial mass (tumor or abscess)	Positional headaches (awakening at night); Cushing triad (hypertension, bradycardia, and irregular respirations); blurry vision, personality changes	ICP crises, herniation
Spinal cord pathology (nontraumatic)	Neck pain or back pain; areflexia or hyperreflexia; saddle anesthesia; bladder and bowel dysfunction; erectile dysfunction	Persistent paralysis and disability; acute respiratory failure; hemodynamic instability
Acute inflammatory demyelinating polyradiculoneuropathy (AIDP), acute motor axonal neuropathy (AMAN)	Progressive ascending weakness or numbness; changes in voice (sounding hoarse, softer, thicker); diplopia, blurry vision, gaze palsies; dysarthria and dysphagia	Hypercarbic respiratory failure, acidosis
Myasthenia gravis	Weakness, worse with exertion and worse toward the end of the day; diplopia, blurry vision, gaze palsies; ptosis; impaired neck strength	Hypercarbic respiratory failure, acidosis

stimulator should be performed to assess for residual neuromuscular blockade, and if there is concern for ongoing paralysis based on the number of twitches, reversal may be considered (if the examination is necessary for time-sensitive decisions) or the evaluation will have to be delayed.

Effects of metabolic and endocrine derangements, infections, hypothermia, and other forms of systemic illness should be factored into the assessment. Last, patients with acute neurologic emergencies frequently present with severe concomitant systemic illness and injuries (eg, polytrauma or refractory shock due to neurogenic cardiomyopathy). In addition to the neurologic examination, hemodynamic assessments and examination of other relevant organ systems are an essential part of the evaluation. Patients with altered consciousness or spinal cord injuries may not reliably report pain or other symptoms related to extracerebral injuries, so a thorough head-to-toe examination with attention to trauma, musculoskeletal, and vascular findings should be carefully performed. Additionally, procedures to treat or temporize neurologic emergencies may precipitate non-neurologic complications. For example, patients undergoing diagnostic cerebral angiography after receiving thrombolysis should be assessed for access site (eg, groin, wrist) hematomas.²³

The neurologic examination is usually tailored to the part of the neuraxis that is believed to be affected. In general, the differentiation between CNS and PNS pathology can be quickly made based on the presence of altered consciousness and corticospinal pathology (CNS), as well as upper (CNS) versus lower (PNS) motor neuron findings based on tone, reflexes, presence of atrophy and fasciculations, and pattern of weakness. It is important to note that CNS and PNS pathologies can coexist and that chronic pathologies such as cervical spondylosis or diabetic neuropathy may confound the examination. Notably, there are acute CNS and PNS pathologies that can closely mimic each other, such as acute inflammatory demyelinating polyradiculoneuropathy (AIDP) with involvement of cranial nerves, or acute spinal pathology that initially presents with areflexia (**CASE 1-2**). Usually, a combination of focused history and additional tests can further differentiate these cases.

Central Nervous System

The most common neurologic emergencies affecting the CNS include various subtypes of severe acute brain injury, traumatic spinal cord injury, and other rapidly progressive spinal cord pathologies.

ACUTE BRAIN INJURY. In patients with severe acute brain injury, a key priority of the examination is to identify signs of elevated ICP and herniation (**FIGURE 1-4**²⁴), which potentially require emergent decompressive craniectomy, hematoma evacuation, an extraventricular drain, or medical management; for more information, refer to the article “Neurocritical Care for Patients With Ischemic Stroke” by T. M. Leslie-Mazwi, MD,²⁵ as well as “Emergent Management of Intracerebral Hemorrhage” by Santosh B. Murthy, MD, MPH, FNCS,²⁶ and “Traumatic Brain Injury and Traumatic Spinal Cord Injury” by Jamie E. Podell, MD, and Nicholas A. Morris, MD, FAAN, FNCS,²⁷ in this issue of *Continuum*.

Another critical early step is to assess for patterns concerning for acute stroke syndromes based on vascular territories, with special attention to brainstem and cerebellar pathology, while considering indications and contraindications for

KEY POINTS

- The neurologic examination may rapidly change in patients with acute neurologic emergencies, and the potential for deterioration must be identified to determine the optimal frequency of evaluations.
- The clinical examination of patients in neurocritical care settings is often limited because of impaired level of consciousness, and confounders such as sedation, neuromuscular blockade, and toxic-metabolic and infectious processes must be considered.

thrombolysis and endovascular mechanical thrombectomy. If there is concern for decreased or fluctuating mental status, a quick assessment for signs of recent (bitten tongue, incontinence) or ongoing subtle (gaze deviation, head-turning, eye-fluttering, chewing on the endotracheal tube) seizure activity is warranted to inform decisions about antiseizure medications and urgent EEG. Evaluating for

CASE 1-1

A 45-year-old man was admitted to the hospital after a high-speed motor vehicle crash. His initial Glasgow Coma Scale score was 15, and initial imaging studies revealed small amounts of scattered, bilateral traumatic subarachnoid hemorrhage and injury of the left vertebral artery. On hospital day two, he reported dizziness, nausea, and headaches. Urgent head CT did not reveal any changes. He was treated with pain medicine. When attempting to work with physical therapy, he was noted to stumble toward the right side. His symptoms persisted over the next 24 hours, and neurologic examination revealed direction-changing nystagmus on horizontal gaze. A repeat CT showed a large ischemic cerebellar infarct involving the posterior inferior cerebellar artery territory (FIGURE 1-1A and 1-1B) with effacement of the fourth ventricle. He was transferred to the neurocritical care unit, and neurosurgery was consulted. Over the next few hours, he became obtunded (Glasgow Coma Scale score 8), and repeat CT showed evolving hydrocephalus (FIGURE 1-1C). An extraventricular drain was placed and set at 20 cm H₂O, and emergent suboccipital craniectomy was performed. Postoperatively, the extraventricular drain was lowered to 10 cm H₂O. At about 24 to 48 hours after surgery, the patient was alert, was appropriately interactive, and followed commands. He reported severe vertigo and nausea and kept his eyes closed preferentially. At follow-up in the clinic 6 months later, he had no neurologic deficits and returned to work as an engineer.

COMMENT

Patients with traumatic brain injury can develop ischemic strokes due to blunt cerebrovascular injury. CT of the head can be negative in the first hours following ischemic stroke; if there is clinical suspicion for an infarct, urgent MRI (if available) or repeat CT should be considered. Strokes affecting the posterior circulation can be subtle, making diagnosis more challenging. Examination of gait is important to identify cerebellar pathology because lesions predominantly affecting the vermis may manifest with only truncal ataxia and no appendicular signs. Cerebellar infarcts can be life-threatening because of brainstem compression and obliteration of the fourth ventricle resulting in obstructive hydrocephalus. With early recognition and prompt treatment, patients are expected to have an excellent long-term prognosis. Aggressive CSF diversion before suboccipital craniectomy can cause upward herniation, but this risk can be mitigated by keeping the extraventricular drain at 20 cm H₂O to only allow for sporadic drainage with high intracranial pressure until decompressive surgery is performed.

meningismus, photophobia, or other signs of systemic infection (eg, stigmata of endocarditis) can help solidify the suspicion of an infection involving the CNS and the need for a prompt LP.

The detailed examination of patients with acute brain injury often must be tailored to the level of consciousness. Wakefulness and awareness should be

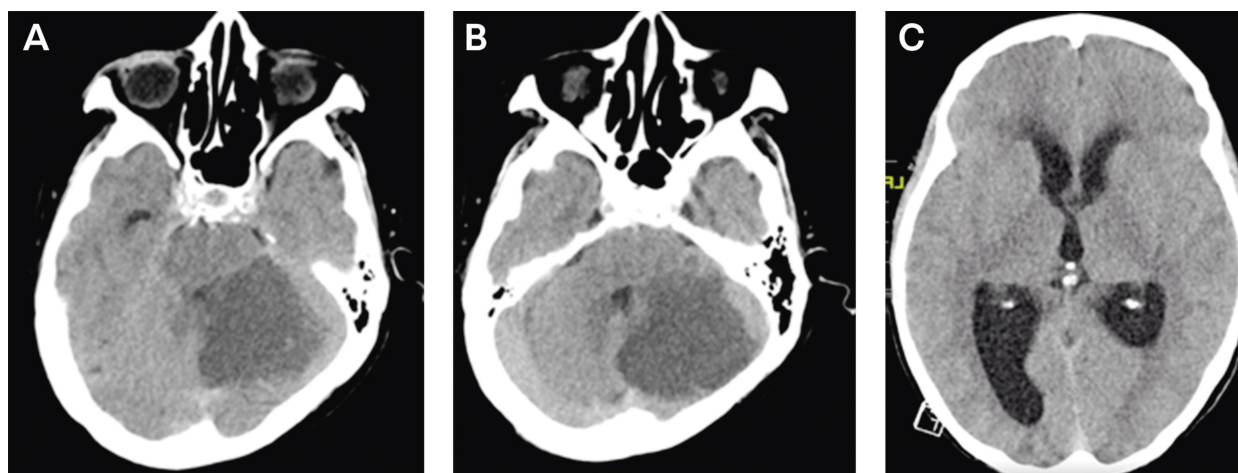
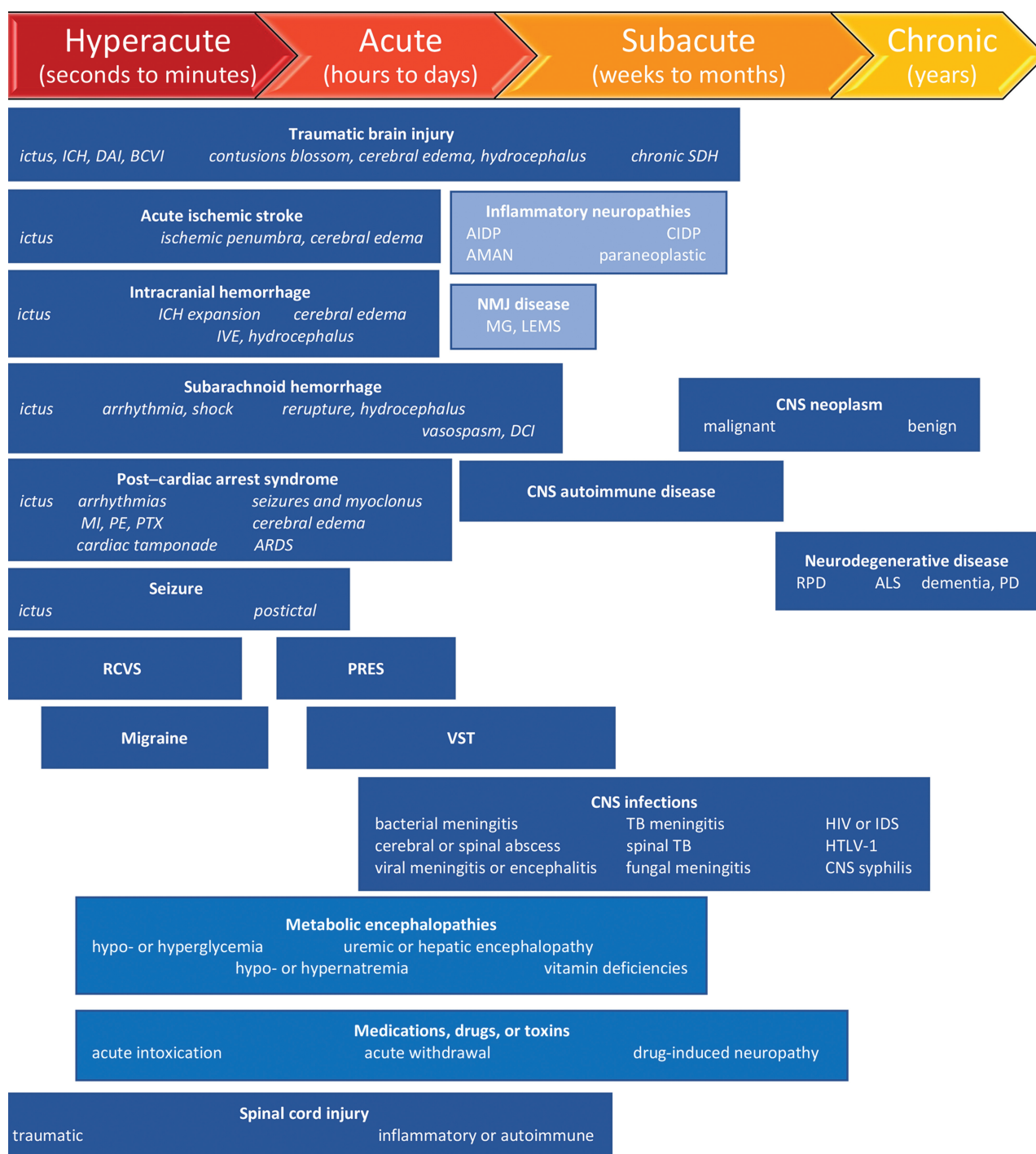


FIGURE 1-1

Imaging from the patient in **CASE 1-1**. Axial noncontrast head CT of a patient with a left-sided cerebellar infarct, causing mass effect on the brainstem and effacing the fourth ventricle (**A, B**) and resulting in evolving hydrocephalus (**C**).

Courtesy of Thuhien Nguyen, MD.

**FIGURE 1-2**

Time course of neurologic diagnoses. Although there are no sharp boundaries in the time course of diagnostic groups, the temporal profile is an essential diagnostic component. Dark blue boxes reflect CNS pathologies, light blue boxes contain PNS pathologies, and the medium blue boxes show pathologies that affect both CNS and PNS.

ALS = amyotrophic lateral sclerosis; AIDP = acute inflammatory demyelinating polyneuropathy; AIDS = acquired immunodeficiency syndrome; AMAN = acute motor axonal neuropathy; ARDS = acute respiratory distress syndrome; BCVI = blunt cerebrovascular injury; CIDP = chronic inflammatory demyelinating polyneuropathy; CNS = central nervous system; DAI = diffuse axonal injury; DCI = delayed cerebral ischemia; HIV = human immunodeficiency virus; HTLV-1 = human T-lymphotropic virus 1; ICH = intracranial hemorrhage; IVE = intraventricular extension; LEMS = Lambert-Eaton myasthenic syndrome; MG = myasthenia gravis; MI = myocardial infarction; NMJ = neuromuscular junction; PD = Parkinson disease; PE = pulmonary embolism; PNS = peripheral nervous system; PRES = posterior reversible encephalopathy syndrome; PTX = pneumothorax; RCVS = reversible cerebral vasoconstriction syndrome; RPD = rapid progressive dementia; SDH = subdural hemorrhage; TB = tuberculosis; VST = venous sinus thrombosis.

assessed first to gauge if the patient is conscious. Generally, responses to vocal and tactile stimulation should be tested before escalating to noxious stimulation. Before evaluating for disorders of consciousness, reversible confounders (intoxication, sedation or paralysis, metabolic derangements) should be eliminated as much as possible. Coma mimics (eg, locked-in syndrome due to basilar artery occlusion, AIDP variants with multiple cranial nerve involvement) should be considered because the patient might be fully conscious and cognizant of conversations at the bedside. In a comatose patient, key portions of the neurologic examination are assessing eye position, examining brainstem reflexes, and checking motor responses to noxious stimuli.

Gaze deviations typically indicate pathology involving the frontal eye fields (bilateral horizontal deviation, ipsilateral to the side of a lesion or postictal state and contralateral to a hemisphere with ongoing seizures), thalamus (bilateral downward deviation, often due to large thalamic ICH), midbrain (Parinaud syndrome [upgaze paralysis due to tumor or hydrocephalus], down and out ocular deviation due to injury or compression of the third cranial nerve nucleus), or pons (vertical or skew deviation) but can also be seen in the context of extensive bihemispheric cortical injury (eg, upward deviation due to diffuse hypoxic ischemic encephalopathy after cardiac arrest). A doll's eye response (ie, the eyes move in the opposite direction of the passively turned head) in comatose patients indicates intact cranial nerve nuclei, whereas an absent response suggests brainstem dysfunction or conscious suppression of the reflex.

Examination of brainstem reflexes includes the blink reflex, pupillary reflex, corneal reflex, oculoccephalic reflex, oculovestibular reflex, gag reflex, and cough reflex. The results of this evaluation can be used to determine the level and extent of brainstem injury and be factored into clinical decisions such as whether to pursue neurosurgical interventions or consider extubation (with intact gag and cough reflexes suggesting better ability to manage secretions).^{28,29} Serial assessment of pupillary function is one of the most important components of the examination of comatose patients because it is an important marker of mass effect or injury to the midbrain. Automated pupillometry can provide quantifiable information about pupillary size and reactivity and is increasingly used in the neurocritical care setting.^{30,31}

Motor responses in unconscious patients are graded on a continuum from stimulus localization to no response. Although responses can be categorized based on the Glasgow Come Scale (GCS), they are best communicated and documented in a descriptive manner rather than using a number on the scale. Change in motor response is often the first sign of acute deterioration. Movements in response to noxious stimulation include decorticate (slow flexion of the elbow and wrist, grasping of the fingers) or decerebrate (adduction and internal rotation of the shoulder, arm extension, and wrist pronation forming a fist) responses, which may help identify the level of cerebral dysfunction.

A variety of abnormal movements can be observed in patients with acute brain injury, including seizures, myoclonus, tremor, and shivering or rigors. The first step in teasing out the significance of these movements is to describe them. Are they regular (seizure, tremor) or irregular (myoclonus, shivering)? Are they diffuse (shivering, myoclonus, or generalized seizure) or focal (some forms of myoclonus, focal seizure)? Which body parts are affected? What are the amplitude and frequency, and do they occur predominantly at rest (parkinsonian

KEY POINTS

- In patients with severe acute brain injury, key priorities of the examination are to identify signs of elevated intracranial pressure and herniation, acute neurovascular syndromes, clinical or subclinical seizures, and central nervous system infection, as each of these pathologies requires different types of workup and emergent treatment.

- The detailed examination of patients with acute brain injury often must be tailored to the level of consciousness. Wakefulness and awareness should be assessed first to gauge if the patient is conscious.

- Key portions of the neurologic examination of comatose patients include assessing eye position, examining brainstem reflexes, and checking motor responses to noxious stimuli. Coma mimics such as locked-in syndrome should be considered.

- In patients with suspected brain death, a list of prerequisites must be met and carefully evaluated before initiating the formal brain death examination.

tremor), or are they enhanced with certain postures (postural tremor or myoclonus) or movements (intention tremor)? Are they induced by tactile stimuli (some forms of myoclonus)? Do they occur in the context of temperature management or acute infection (rigor), or in the presence of metabolic derangements due to liver or kidney failure (myoclonus)? Associated findings such as altered mental status, gaze deviation, head-turning, lip-smacking, and tonic-clonic posturing should heighten the concern for seizure. The ability to follow commands, track, or communicate while the movements occur lowers the suspicion for generalized seizures. In critically ill patients, different types of adventitious movements can occur simultaneously and are often hard to distinguish. EEG monitoring is often indicated to diagnose or exclude electrographic seizures in unresponsive patients with a limited clinical examination and high risk for seizure.

In patients with devastating brain injury, persistent coma, loss of all brainstem reflexes, and the inability to breathe spontaneously, the possibility of brain death or death by neurologic criteria must be considered. A thorough description of the brain death evaluation is beyond the scope of this review. However, it is

CASE 1-2

A 52-year-old woman with a history of diabetes presented to the emergency department with malaise and acute weakness in both lower and upper extremities. She reported that the symptoms started 2 days before with tingling in her feet, which progressed to her arms and legs. Her neurologic examination was notable for diffuse motor weakness in her arms and legs (mostly 2/5 to 3/5 with no length-dependent pattern), with 1/5 shoulder abduction, and absent muscle stretch reflexes throughout. She was started on intravenous immunoglobulin (IVIg) and admitted to the neurocritical care unit. Ten hours after admission, her deficits progressed to quadriplegia. Her examination was notable for midline neck pain. Her vital signs were notable for hypotension (blood pressure 80/50 mm Hg) and bradycardia (heart rate 40 beats/min to 50 beats/min), and her laboratory values were notable for leukocytosis. Her breathing was rapid and shallow, and her maximum inspiratory pressure was 20 cm H₂O, so she was intubated. MRI of her spine showed a cervical epidural abscess extending from C2 to T2 (FIGURE 1-3), and she underwent emergent surgical decompression and drainage. Previous blood cultures and cultures from the surgical site returned positive for methicillin-resistant *Staphylococcus aureus*, and she was started on a long-term course of vancomycin. After a prolonged hospital course, which included placement of a tracheostomy, she was discharged to rehabilitation. A year later, she continued to require a wheelchair but regained some strength in her upper and lower extremities.

important to note that, before initiating a formal brain death evaluation, a series of prerequisites must be met.³²⁻³⁴ The clinical history, etiology, and neuroimaging should be consistent with a permanent catastrophic injury to the brain as a whole, and potentially confounding factors should be excluded (eg, sedating medications, hypothermia, profound metabolic or endocrine derangements, hypotension).

ACUTE SPINAL CORD INJURY. Evaluation of suspected spinal cord injury typically starts with assessing the level and type of injury based on patterns of motor and sensory function. The constellation of motor and sensory symptoms and the nature of sensory deficits can provide important clues about the affected tracts, level, and underlying pathology. Reduced or absent rectal tone can be an early clue for spinal cord pathology. The presence of autonomic dysfunction (impaired cardiovascular and hemodynamic regulation, loss of bladder and bowel control) provides additional information regarding localization and may guide further management. Although muscle tone and reflexes are typically increased as a

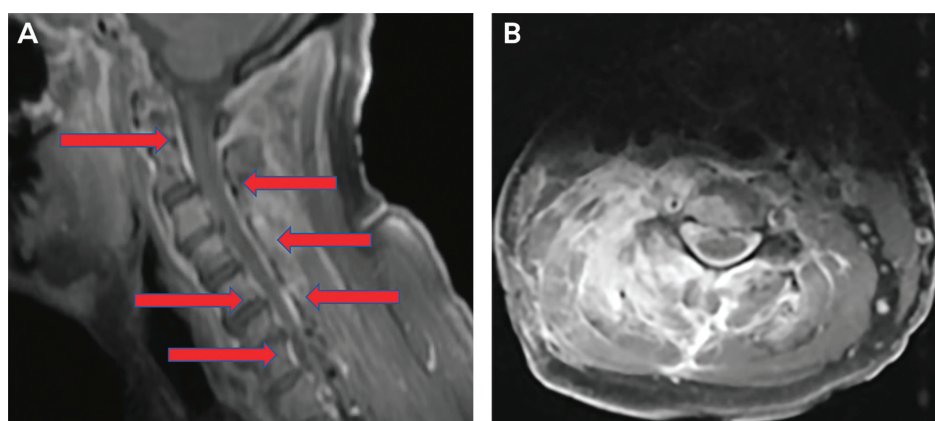
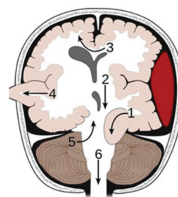


FIGURE 1-3
Imaging from the patient in **CASE 1-2**. Sagittal T1-weighted postcontrast (A) and axial T1-weighted postcontrast (B) MRI. Enhancement in the epidural space traversing from C2 to T2 (A, arrows), concerning for a phlegmon, and bulging discs at C3 to C5, resulting in anterior cord compression.

Acute spinal cord pathology can closely mimic a peripheral nervous system process such as Guillain-Barré syndrome. Muscle stretch reflexes can be absent in the acute phase because of spinal shock. In patients presenting with acute progressive quadriparesis or quadriplegia, the pattern of weakness and sensation are important tools to determine localization. Although neck pain, back pain, and hemodynamic instability can also be encountered in Guillain-Barré syndrome, these symptoms should prompt clinicians to exclude a spinal process. Patients with spinal cord lesions at C5 or higher are at risk for acute respiratory failure even if there is no clear evidence of respiratory distress due to neuromuscular weakness.

COMMENT



Type of herniation	Neuroanatomical pathology	Clinical signs
1 Uncal or lateral transtentorial herniation	Temporal lobe medializes toward and below the tentorium causing midbrain and third nerve compression; posterior cerebral artery (PCA) compression; may result in an indentation of the contralateral cerebral peduncle against the tentorium (Kernohan notch)	Ipsilateral fixed and dilated pupil, contralateral hemiparesis, coma; PCA syndrome; ipsilateral hemiparesis with Kernohan phenomenon
2 Central transtentorial herniation	Downward displacement of the diencephalon and brainstem causing compression of the hypothalamus and reticular activating system	Coma, bilateral small pupils; decorticate then decerebrate posturing; rostral then caudal loss of brainstem reflexes; Parinaud syndrome (loss of upgaze, retraction nystagmus); diabetes insipidus
3 Subfalcine or cingulate herniation	Displacement of the cingulate gyrus, corpus callosum (and ipsilateral anterior cerebral artery [ACA] or pericallosal artery) under the falx; compression of the contralateral foramen of Monro	Coma; ACA syndrome (contralateral leg weakness); dilation of the contralateral ventricle
4 Transcalvarial herniation	Cerebral displacement through fracture or surgical skull defect	Based on the location of the skull defect and underlying cortex
5 Cerebellar upward or ascending transtentorial herniation	Increased pressure in the posterior fossa, pushing the midbrain upward through the tentorial notch; PCA and superior cerebellar artery (SCA) compression	Cerebellar dysfunction, pupillary abnormalities; hydrocephalus (anterior brainstem displacement); PCA and SCA syndrome
6 Cerebellar downward or tonsillar herniation	Cerebellar tonsils move downward via the foramen magnum causing brainstem, cerebellar, fourth ventricular compression; posterior inferior cerebellar artery compression	Head tilt, neck stiffness, cerebellar dysfunction; severe headaches (if conscious) followed by coma; hemodynamic instability (often hypertension); cardiorespiratory dysfunction, cardiac arrest; Wallenberg syndrome

FIGURE 1-4
Herniation syndromes.

Illustration reprinted from Life in the Fastlane.²⁴ © 2024 LITFL.

result of spinal cord injury, there is often an initial period of spinal shock with loss of muscle tone and areflexia in the acute period.

Lesions with a spinal level at C5 or above, especially complete lesions, can result in weakness of the diaphragm and other respiratory muscles, as well as hypoventilation and pulmonary atelectasis due to loss of spinal baroreceptors, requiring mechanical ventilation.³⁵ Lesions at or higher than the T6 level can manifest with loss of sympathetic innervation and vasomotor tone, resulting in peripheral vasodilatation and inability to produce reflexive tachycardia; this combination can result in life-threatening hemodynamic compromise.³⁶

CASE 1-3

A 39-year-old woman was brought to the hospital after a fall from a horse. Her initial Glasgow Coma Scale (GCS) score in the emergency department was 10 (eye response: E3; motor response: M4; visual response: V3), and her head CT was notable for bifrontal contusions and a small right-sided subdural hematoma. Over the next few hours, her GCS score deteriorated to 7 (E1; M3; V2). She displayed loud snoring sounds and pooling of secretions and was intubated for airway protection. A repeat CT revealed a slight increase in contusion size and effacement of the basal cisterns. She was admitted to the intensive care unit and noted to be bradycardic (heart rate 40 to 50 beats/min) and hypertensive (blood pressure 170/90 mm Hg). She was given a dose of mannitol, and her heart rate normalized at 70 to 80 beats/min. An intraparenchymal intracranial pressure (ICP) monitor was placed and demonstrated an initial ICP of 35 cm H₂O. She received additional hyperosmolar therapy. A repeat CT was stable, although her ICP continued to be elevated. She was started on a propofol infusion. Over the next 24 hours, she received three doses of hypertonic saline for transiently elevated ICP.

On hospital day 2, she was bradycardic, with decreased pupillary reactivity and persistently elevated ICP; her GCS score deteriorated to 5. A repeat CT showed blossoming of the contusions (FIGURE 1-5), and an emergent bifrontal craniectomy was performed with subsequent improvement of ICP, pupillary reactivity, and mass effect on CT. Her ICP monitor was removed 24 hours later. Brain MRI revealed a bilateral grade 1 diffuse axonal injury. Over the next week, she became more alert. After a prolonged rehabilitation course, she returned to work as a librarian and had mild persistent cognitive deficits 1 year after her injury.



FIGURE 1-5
Imaging from the patient in CASE 1-3. Axial noncontrast head CT with bifrontal contusions and traumatic subarachnoid hemorrhage.

In patients with severe traumatic brain injury, an ICP monitor should be considered based on neurologic examination (GCS score less than 8, pupillary changes, clinical findings suggestive of intracranial hypertension such as concomitant hypertension and bradycardia) and CT findings (appearance of sulci and cisterns, edema, absence of mass lesion). ICP is typically managed in a tiered approach with a stepwise escalation.

COMMENT

Peripheral Nervous System

In acute polyradiculoneuropathies (eg, AIDP or acute motor axonal neuropathy [AMAN]) and disorders of the neuromuscular junction (eg, myasthenia gravis, botulism), a multitude of clinical examination findings can be encountered. The most critical task is to gauge the potential for neuromuscular respiratory failure and need for tracheal intubation. A general rule is that any patient with neuromuscular weakness and signs of respiratory distress, including tachypnea, increased work of breathing, diaphoresis, paradoxical breathing patterns, and use of accessory respiratory muscles, should be intubated. Patients who appear well can be monitored by using objective respiratory measurements, including vital capacity (the total amount of air exhaled after maximal inhalation), maximum inspiratory pressure (patients are asked to exhale and then perform a maximal inspiratory effort, sustaining it for 1 to 2 seconds) and maximum expiratory pressure (patients are asked to perform a maximal expiratory effort and sustain it for 1 to 2 seconds). Trends in these measurements and the 20/30/40 rule (vital capacity less than 20 mL/kg, maximum inspiratory pressure worse than -30 cm H₂O, maximum expiratory pressure less than 40 cm H₂O) can be used to guide intubation decisions, typically in combination with the patient's clinical appearance. The single-breath count test (the patient inhales deeply while sitting upright, then counts approximately two numbers per second out loud to as high as possible until the next breath) is also thought to correlate with vital capacity³⁷ and can serve as good bedside evaluation of respiratory mechanics that can be conducted and followed by any member of the treatment team. However, all these parameters may be confounded by limited effort. Arterial blood gas abnormalities, namely respiratory acidosis, are thought to occur in the late stages of neuromuscular respiratory failure (at a time when a patient should have already been intubated), so a normal arterial blood gas must be interpreted in the context of the clinical picture.

The Erasmus Guillain-Barré Syndrome Respiratory Insufficiency Score is a prediction model to gauge the probability of acute respiratory failure within the first week of intubation in patients with AIDP. The main predictors of intubation include the length of time between the onset of weakness and hospital admission, facial and bulbar weakness on admission, and the Medical Research Council muscle strength sum score.³⁸ Other early predictors of the need for mechanical ventilation are an inability to stand, cough, or lift the elbows or head.³⁹ Patients with extensive cranial nerve impairment, especially bifacial weakness and rapid changes in voice and swallowing function, warrant close monitoring. Assessing the rate of progression of motor weakness, sensory symptoms, and areflexia can provide clues about the severity and pace of the disease process. In addition, patients with AIDP and AMAN can present with profound autonomic instability; therefore, vital signs should be assessed and monitored closely.

Patients with myasthenia gravis may not exhibit typical signs of dyspnea because of profound generalized weakness. Given the fluctuating nature of symptoms with this disorder, respiratory parameters such as vital capacity, maximum inspiratory pressure, and maximum expiratory pressure may not always reflect the patient's trajectory, and these measurements have to be complemented by close monitoring of overall strength and fatigability and ocular motor and bulbar symptoms. Diffuse fatigue, worsening neck flexion and extension, and voice changes can herald the need for intubation. Prominent bulbar symptoms often result in an inability to manage secretions, and

anticholinesterase medications may contribute to hypersalivation, increased secretions, and worsening weakness; for more information on the management of neuromuscular respiratory failure, refer to the article "Neuromuscular Emergencies" by Catherine S. W. Ablin, MD,⁴⁰ in this issue of *Continuum*.

DETECTING DETERIORATION

A critical aspect of the initial evaluation in neurocritical care is to determine the potential for subsequent deterioration and the need for close monitoring to minimize the risk of secondary brain injury. The causes of neurologic or systemic worsening are related to the disease process (**TABLE 1-1**), and there are critical time windows for different pathologies (**FIGURE 1-1**). The major types of deterioration in patients in neurocritical care include (1) ICP crises due to bleeding (rebleed or expansion; **CASE 1-3**), edema of brain tissue (cytotoxic or vasogenic), or disruption of CSF flow or absorption (hydrocephalus); (2) impaired cerebral perfusion (due to vascular stenosis, occlusion, vasospasm, or systemic hypoperfusion); (3) prolonged generalized seizures, causing imbalance in metabolic demand and supply; (4) acute respiratory failure due to insufficient airway protection, impaired respiratory drive, or neuromuscular weakness; or (5) systemic complications resulting in hemodynamic instability. The risk of deterioration can typically be gauged based on a combination of the examination (vital signs, level of consciousness, pupillary examination, presence or absence of posturing, signs of elevated ICP, airway protection, oxygenation and ventilation, hemodynamic assessment) and neuroimaging studies (size of a mass lesion, degree of midline shift and mass effect, impending herniation, risk of obstructive hydrocephalus, presence of proximal arterial occlusion, or stenosis of the intracranial or cervical vasculature).

Changes in the neurologic examination are often the first sign of an acute process, warranting urgent management. Ideally, the clinical examination of patients with a high risk of deterioration should be monitored in an intensive care unit by clinicians trained in neurocritical care and experienced with this population.¹⁰ When determining the frequency of checks, the potential for immediate worsening must be balanced with concerns for delirium and agitation.⁴¹ When unable to follow the clinical examination, additional monitoring with serial imaging, EEG, invasive and noninvasive ICP monitoring, and other modalities may be indicated (**TABLE 1-3**).

SCALES AND SCORES

A multitude of clinical and radiographic scales and scores are used to assess and track patients in neurocritical care (**TABLE 1-4**⁴²⁻⁶¹) and the utility and significance of these vary. Often, scales quantify initial disease severity based on relevant clinical features and provide an objective measure to consistently track the evolution of symptoms over time between different examiners. This quantification allows for categorization, stratification, and measurement of improvement or worsening in clinical trials. In some instances, scales are used to guide clinical decision making (eg, the GCS and Marshall CT classification scores are often used to consider ICP monitor placement,⁶² and the National Institutes of Health Stroke Scale and Alberta Stroke Program Early CT scores are key parts of the evaluation for eligibility for thrombolysis or mechanical thrombectomy in patients with acute ischemic stroke).⁶³ Scales and scores may assist in quantifying disease severity and gauging the prognosis, but as is discussed in the article

KEY POINTS

- Although muscle tone and reflexes are typically increased as a result of spinal cord injury, there is often an initial period of spinal shock with loss of muscle tone and areflexia in the acute period.
- In patients with acute presentations of spinal cord injury, polyradiculoneuropathies, or neuromuscular junction disorders, assessing respiratory function and monitoring for signs of acute neuromuscular respiratory failure are critical parts of the examination.
- Patients in neurocritical care settings are often at risk for subsequent deterioration due to intracranial pressure crises, impaired cerebral perfusion, prolonged seizures and status epilepticus, acute respiratory failure, and hemodynamic instability. Commonly used monitoring strategies include serial neurologic examination, pupillometry, serial imaging, invasive and noninvasive intracranial pressure monitoring, and EEG monitoring.

“Prognostication in Neurocritical Care” by Susanne Muehlschlegel, MD, MPH, FNCS, FCCM, FAAN,⁶⁴ in this issue of *Continuum*, studies on their relationship with outcome may be biased by nihilism and the self-fulfilling prophecy because of withdrawal of life-sustaining treatments.⁶⁵

Scales and scores must also be cautiously interpreted in the context of clinical confounders. When making treatment decisions or prognostic assessments, these metrics should always be used in conjunction with a nuanced, detailed neurologic examination and consideration of imaging results, as well as the overall clinical context. For example, the GCS score may be falsely low for an aphasic patient (losing points on the motor scale for an inability to follow commands and on the verbal scale because their language impairment may mimic confusion) or an alert

TABLE 1-3 Summary of Monitoring Modalities in Neurocritical Care

Monitoring modality	Primary signal output	Indications and applications
Noninvasive		
Pupillometry	Pupillary size; neurologic pupillary index	Concerns for neurologic deterioration due to risk of elevated intracranial pressure, brain death clinical examination
Transcranial Doppler	Cerebral blood flow velocity; peak systolic velocity; end diastolic velocity; mean flow velocity; Lindegaard ratio (ratio of the highest mean velocities in the middle cerebral artery and the ipsilateral extracranial internal carotid artery); pulsatility index	Vasospasm monitoring, detection of stenosis and acute cerebrovascular occlusions, assessment of direction of flow and collateral flow, microemboli detection, assessment of cerebral autoregulation and vasoreactivity, bubble study to assess for intracardiac or intrapulmonary shunt, screening for angiopathy in sickle cell disease, ancillary study in brain death determination, confirm cerebral circulatory arrest
Quantitative electroencephalography (EEG)	Power at different frequencies creating a compressed spectral array	Objective and efficient EEG review, bedside screening of EEG data; detection of asymmetries; monitoring of burst suppression; detection of nonconvulsive seizures and status epilepticus; sedation monitoring; vasospasm monitoring
Near-infrared spectroscopy	Regional cerebral oxygen saturation; tissue oxygenation index	Description of cerebral oxygenation and blood flow, assessment of cerebral autoregulation
Invasive		
Intracranial pressure monitoring	Intracranial pressure, cerebral perfusion pressure	Concerns for neurologic deterioration due to risk of elevated intracranial pressure (typically indicated when unable to follow the clinical examination)
Brain tissue oxygen monitoring	Pbto ₂ (partial pressure of oxygen in brain tissue)	Assessment of cerebral oxygen-carrying capacity, detection of brain tissue hypoxia or increased cerebral oxygenation demand
Cerebral metabolism monitoring	Glucose, lactate, pyruvate, neurotransmitters, cytokines	Direct measurement of brain metabolism and toxic metabolites, limited evidence

patient with eyelid-opening apraxia. The ICH Score may incorrectly predict poor outcomes in a patient with a large ICH in the nondominant parietooccipital lobe or relatively minor intraventricular hemorrhage.

IMAGING

Imaging studies complement the neurologic examination to identify structural lesions and guide management. In general, CT identifies most emergent pathologies, while MRI provides more nuanced information that can inform workup and prognostication.

Central Nervous System

Imaging studies are often an essential part of the workup for neurologic emergencies affecting the CNS because identifying structural pathologies in patients with acute brain and spinal cord injuries determines the indication for urgent neurosurgical intervention.

ACUTE BRAIN INJURY. Any new, unexplained deterioration in the patient's level of consciousness or a new focal neurologic deficit warrants emergent CT of the head. Noncontrast CT of the head is a relatively fast study that reliably identifies structural abnormalities within the brain parenchyma, ventricular system, and bone. A structured approach when screening a CT of the head for urgent pathologies consists of (1) ensuring appropriate windowing; (2) screening of bony structures for fractures; (3) evaluating for deviation of midline structures and quantifying shift; and (3) assessing for the "4Hs": hyperdensity, hypodensity, hydrocephalus, and herniation.

Hyperdensities are concerning for acute hemorrhage and can be differentiated from calcifications by measuring the Hounsfield units (60 to 100 Hounsfield units for acute hemorrhage, greater than 100 Hounsfield units for calcifications). CT is greater than 95% sensitive for detecting acute blood within 24 hours of symptom onset.^{66,67} Hypodensities are typically either the result of cytotoxic edema in the context of acute cerebral ischemia (involving both gray and white matter, following arterial territories when caused by thromboembolic infarcts) or vasogenic edema related to blood-brain barrier breakdown surrounding a mass or hematoma (and typically sparing the gray matter). Of note, acute ischemia usually takes several hours to evolve on CT and may not be visualized on an early CT. Hydrocephalus can be identified and characterized by assessing the size of the ventricular system at all levels (lateral ventricle size, visibility of temporal horns, and third and fourth ventricular dilation or effacement), CSF volume, presence of transependymal flow, and obstructive pathology. In older adult patients or patients with extensive cortical atrophy, measurement of the ventricle-to-brain ratio and comparison with earlier imaging may be needed to differentiate between acute and chronic hydrocephalus. Herniation is evaluated by using the falx, uncus, tentorium, and cerebellar tonsils as anatomical landmarks (**FIGURE 1-4**) and by assessing for lateral or downward shifts and deviations of midline structures.

CT angiography (CTA) is indicated when there is clinical suspicion for acute vascular pathologies, including occlusion or stenosis of the cervical or intracranial vasculature, ruptured aneurysms or vascular malformations, or cerebral vasospasm. Screening for blunt cerebrovascular injury with CTA should occur in trauma patients with high-risk mechanisms (eg, strangulation, hanging,

KEY POINT

● Scales and scores are used in assessing initial neurocritical care disease severity, following progression, measuring outcome measures in clinical trials, guiding clinical decisions, and assessing prognosis. In an individual patient, these metrics must be used in combination with a nuanced examination and consideration of confounding factors.

high-speed motor vehicle crash) or injuries (eg, mandible fractures, LeFort type II or III facial fractures, skull base fractures, cervical spine fractures, major thoracic injuries). In an alert, cooperative patient, the history and clinical examination can determine the need for vessel imaging based on the presence of an ischemic stroke syndrome, substantial head or neck trauma, or sudden-onset severe headache. Basilar artery occlusion should always be considered as a potentially devastating pathology that can present with subtle symptoms. In a patient with impaired consciousness and limited ability to obtain a detailed examination, CTA should be obtained to exclude critical neurovascular pathology. Findings on CTA can determine the need for a catheter angiogram, indications for which include (1) large vessel occlusion deemed to potentially benefit from mechanical thrombectomy; (2) vascular malformation requiring

TABLE 1-4 Clinical and Radiographic Scales and Scores Commonly Used in Neurocritical Care

Scale or score	Population	Components
Clinical examination		
American Spinal Cord Association Impairment Scale ⁴²	Spinal cord injury	Complete or incomplete motor and sensory function
Erasmus Guillain-Barré Syndrome Respiratory Insufficiency Score ³⁸	Guillain-Barré syndrome	Days between weakness and hospitalization, facial or bulbar weakness, Medical Research Council sum score
Full Outline of UnResponsivness (FOUR) score ⁴³	Mixed neurocritical care population	Motor response, eye response; brainstem reflexes, respiration
Glasgow Coma Scale (GCS) ⁴⁴	Described in traumatic brain injury, often applied to other subtypes of acute brain injury	Eye-opening: 4 = spontaneous, 3 = to sound, 2 = to pain, 1 = none Verbal response: 5 = orientated, 4 = confused, 3 = inappropriate, 2 = incomprehensible, 1 = none Motor response: 6 = obeys commands, 5 = localizes, 4 = flexion withdrawal, 3 = abnormal flexion, 2 = extension, 1 = none
Hughes Functional Grading Scale ⁴⁵	Guillain-Barré syndrome	Ability to run or walk, requiring a walker, requiring mechanical ventilation
Hunt and Hess Scale ⁴⁶	Aneurysmal subarachnoid hemorrhage (SAH)	Level of consciousness, headaches, nuchal rigidity, focal neurologic deficits
National Institutes of Health Stroke Scale ^{5,47}	Acute ischemic stroke	Level of consciousness; comprehension; speech; language; gaze; visual symptoms; face, arm, and leg weakness, sensory symptoms; ataxia; extinction
Quantitative myasthenia gravis score ^{48,49}	Myasthenia gravis	Upward (ptosis) and lateral gaze, facial muscles, swallowing, speech, vital capacity, upper extremity strength
World Federation of Neurological Surgeons grading scale ⁵⁰	Aneurysmal SAH	GCS, presence of motor deficit

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further characterization and potential treatment; (3) absence of vascular pathology on CTA but high suspicion based on the clinical history; (4) ICH of undetermined etiology in a young patient; or (5) symptomatic cerebral vasospasm.

CT perfusion (CTP) has been used increasingly to guide decisions about mechanical thrombectomy in acute ischemic stroke.⁶⁸ CTP shows mean transit time, cerebral blood volume, and cerebral blood flow, which can be used to differentiate the infarct core and the ischemic penumbra. In patients with poor cardiac output, atrial fibrillation, or proximal severe arterial stenosis, CTP must be interpreted with caution because decreased blood flow can lead to inaccurate perfusion maps (overestimated mean transit time, underestimated cerebral blood flow) and erroneous diagnosis of extensive ischemia. CTP has also been

CONTINUED FROM PAGE 576

Scale or score	Population	Components
Clinical data and neuroimaging		
Bleeding Assessment Tool ⁵¹	Nontraumatic intracerebral hemorrhage (ICH)	Time between onset and computed tomography (CT), ICH volume, blend sign, hypodensity, recurrent ICH, intraventricular hemorrhage (IVH), anticoagulation
Functional Outcome in Patients With Primary Intracerebral Hemorrhage score ⁵²	Nontraumatic ICH	GCS; age; ICH volume; lobar, deep, infratentorial location; pre-ICH cognitive impairment
ICH Score ⁵³	Nontraumatic ICH	GCS, age, ICH volume, infratentorial origin, IVH
Neuroimaging		
Alberta Stroke Program Early CT ⁵⁴	Acute ischemic stroke	Hypodensities in predefined areas on CT
Marshall CT classification ⁵⁵	Traumatic brain injury	Midline shift, visibility of cisterns, presence of a 25-cm ³ high-density lesion
Modified Fisher scale ^{56,57}	Aneurysmal SAH	Focal versus diffuse SAH, thin versus thick SAH, IVH
Rotterdam CT classification ⁵⁸	Traumatic brain injury	Midline shift, visibility of cisterns, epidural mass lesion, IVH or traumatic SAH
Clinical examination and other data		
Acute Physiology and Chronic Health Evaluation II ⁵⁹	Patients in intensive care	GCS, temperature, mean arterial pressure, heart rate, respiratory rate, oxygenation, pH, bicarbonate concentration, sodium, potassium, creatinine, hematocrit, white blood cell count
Sequential Organ Failure Assessment ^{60,61}	Developed in sepsis, now applied to other intensive care unit populations	GCS, partial pressure of oxygen/fraction of inspired oxygen ratio and mechanical ventilation, platelet count, mean arterial pressure and vasoactive medications, bilirubin, creatinine

used to identify hypoperfused or at-risk areas in the setting of vasospasm; however, optimal perfusion profiles in this setting have not been established.

MRI of the brain can add valuable diagnostic information because of better soft tissue discrimination than CT. It is superior to CT in detecting small infarcts, early infarcts, brainstem pathology, diffuse axonal injury, cortical and subcortical microhemorrhages; characterizing intracranial masses (tumor or abscess); and gauging the timing of acute ischemic stroke or ICH. In acute neurologic emergencies, CT is the initial neuroimaging modality because it is usually much faster and more readily available and does not require clearance for implanted medical devices or metallic foreign bodies. The need for emergent interventions such as treatment of ICP crises or acute vascular occlusions can be identified based on CT or CTA, and MRI should not delay time-sensitive management. Typically, MRI is obtained in the subacute period, after initial stabilization has occurred, to further characterize pathology seen on CT, to evaluate for findings not well visualized on CT, or when CT findings do not fully

CASE 1-4

A 48-year-old man with no significant medical history presented to the emergency department with 5 days of headache and progressive confusion. His wife recounted that he first reported headache and neck pain, had been irritable and acting strangely, and had visual and auditory hallucinations. She reported that he had intermittent fevers in the past 2 days. His examination was notable for neck stiffness, and his mental status fluctuated, ranging from lethargy to stupor. CT of his head was unremarkable. He was started on empiric broad-spectrum antibiotics and acyclovir. A lumbar puncture revealed elevated protein (140 mg/dL), normal glucose, and 90 white blood cells/mm³ with a lymphocytic predominance. He was intubated because of his worsening level of consciousness.

MRI of his brain showed fluid-attenuated inversion recovery (FLAIR) hyperintensities in his right mesiotemporal lobe and bilateral insular cortex (FIGURE 1-6). When he was examined again by the emergency medicine resident, a gaze deviation to the right was noted. He was given IV lorazepam and levetiracetam. EEG demonstrated left-sided temporal seizures (FIGURE 1-7A and 1-7B). On subsequent continuous EEG, periodic epileptiform discharges at 1.5 Hz to 2 Hz were noted (FIGURE 1-7C). His herpes simplex virus polymerase chain reaction (PCR) returned positive. He was treated with acyclovir and three antiseizure medications, and his EEG improved over the next 24 hours. His mental status improved over the following week. One year later, he still had cognitive and behavioral sequelae and was not able to work, but he lived at home with his family and was able to perform activities of daily living independently.

explain the clinical picture. When ordering and interpreting MRI, it is important to recognize the relevant sequences based on the underlying condition and the clinical question.

ACUTE SPINAL CORD INJURY. Patients with acute trauma should be presumed to have an unstable fracture of the cervical spine until further workup is completed. A fully conscious, stable, cooperative patient can be cleared based on clinical assessment. No further imaging is thought to be necessary for patients with no posterior midline tenderness, no focal neurologic deficits, and no distracting injuries in the absence of confusion or intoxication.⁶⁹ In the setting of trauma with impaired neurologic status, a CT of the cervical spine should be obtained to screen for fractures, misalignment of the bony spine, and evidence of cord compression. When cervical pathology is identified on CT, MRI of the spine should be considered on a case-by-case basis, factoring in clinical examination and extent of injury on CT, as well as MRI availability and potential delays in

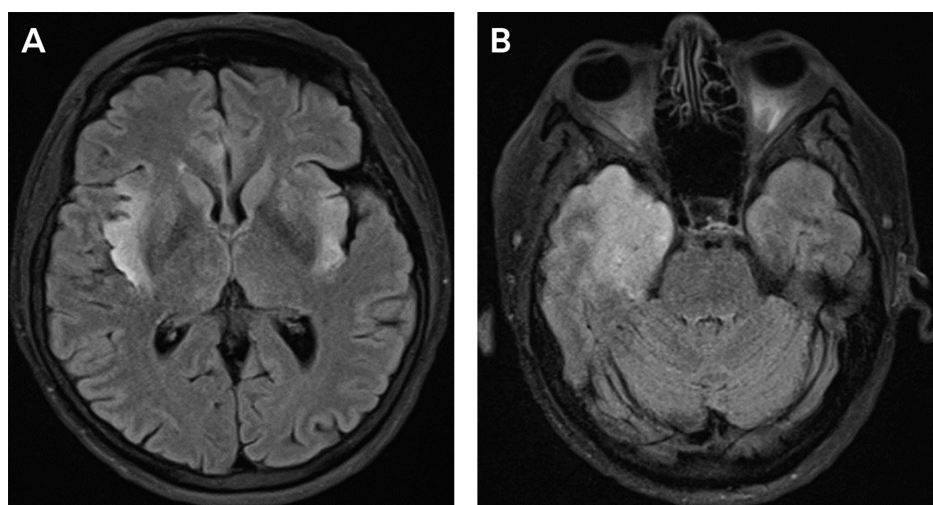


FIGURE 1-6
Imaging from the patient in **CASE 1-4**. Axial fluid-attenuated inversion recovery (FLAIR) images showing bilateral insular (A) and right medial temporal (B) hyperintensities in a patient with herpes simplex virus encephalitis.

Courtesy of Joseph Zunt, MD.

The triad of headaches, fevers, and an acute-subacute onset of confusion should always raise concerns for meningitis or encephalitis. To prevent further neurologic deterioration, broad-spectrum anti-infective agents should be initiated immediately, and lumbar puncture should be performed as soon as possible. Seizures are common in patients with central nervous system infections, and in patients with a decreased or fluctuating level of consciousness, EEG should be obtained to assess for nonconvulsive seizures. Herpes simplex virus encephalitis typically affects the mesiotemporal lobes, which are particularly epileptogenic.

COMMENT

CASE 1-4

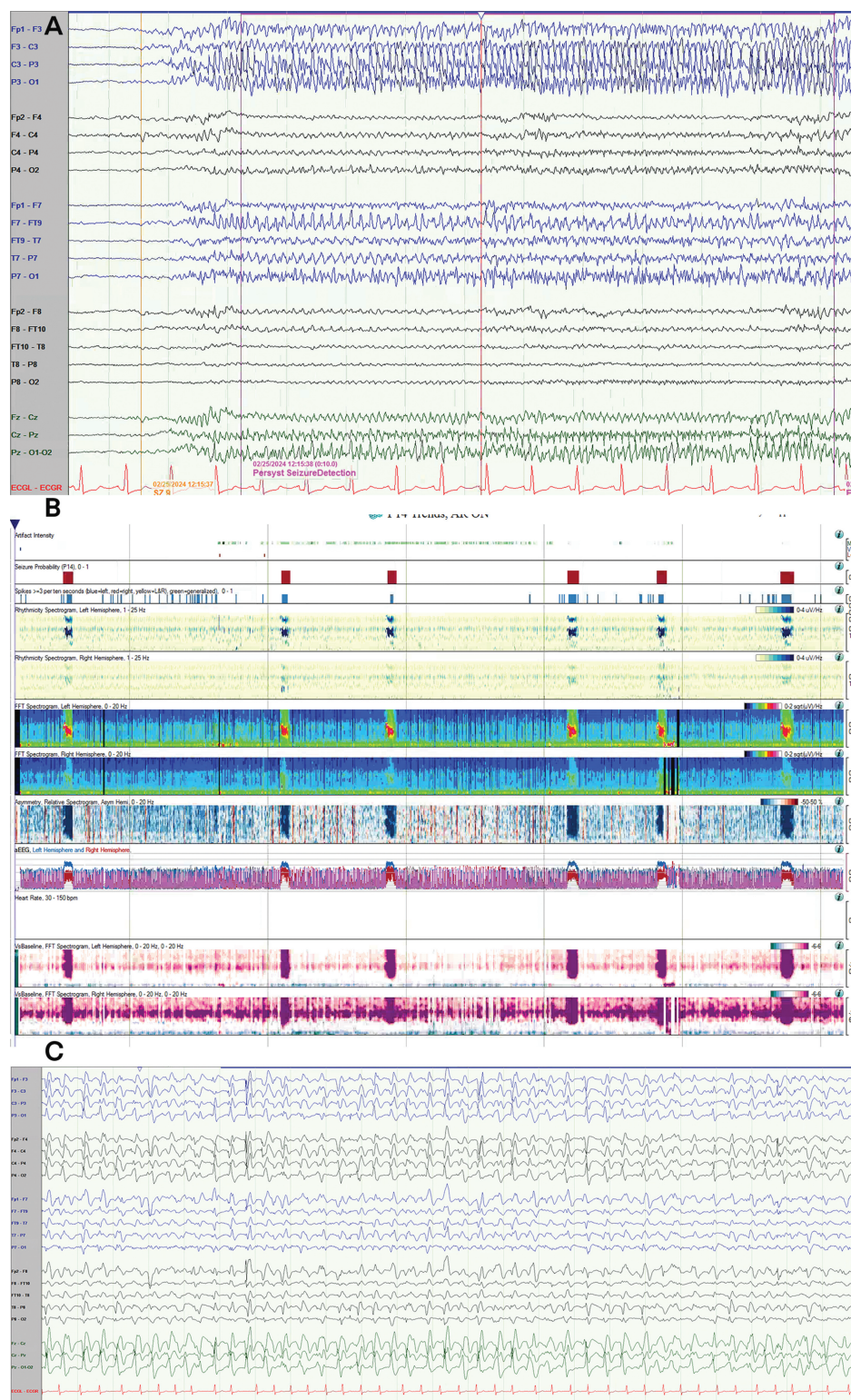
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FIGURE 1-7

EEG findings from the patient in **CASE 1-4**. EEG showing left-hemispheric seizure (A), corresponding quantitative EEG (B), and generalized periodic epileptiform discharges on EEG (C) in a patient with herpes simplex virus encephalitis.

Courtesy of Victor Lin, MD, and Behnaz Esmaeili, MD.

obtaining the MRI. Spinal MRI depicts nuanced details including the extent of intramedullary injury and edema, cord transection, disc and ligamentous injury, intramedullary or epidural hemorrhage, and posttraumatic syringomyelia, which may guide the operative approach. The need for additional spinal imaging of the thoracolumbar cord should be determined based on the trauma mechanism and symptoms.

For patients with suspected nontraumatic spinal cord pathology, MRI of the spine is the best study to characterize infectious or malignant processes within the spinal canal or evaluate autoimmune or inflammatory intramedullary pathology. Because MRI is not always immediately available, an initial CT may be considered to screen for signs of external cord compression or subclinical bony fractures when there are substantial deficits and a high clinical concern.

Peripheral Nervous System

Diagnoses of PNS disorders are usually based on the clinical examination, complemented by additional laboratory workup and electrophysiologic studies. In patients with new flaccid quadriparesis in whom the diagnosis of Guillain-Barré syndrome is in doubt, cervical spine MRI should be considered. For patients with a new diagnosis of myasthenia gravis, CT of the chest should be obtained to assess for the presence of a thymoma, which would impact management.

Systemic Imaging

Patients in neurocritical care settings often need systemic imaging. A detailed review of circumstances that warrant systemic imaging and imaging modalities is outside the scope of this article. However, it is important to note that CT imaging of the chest and abdomen should be obtained in patients with acute trauma to assess for concomitant injuries and in those with a newly diagnosed CNS mass to assess for systemic infection or malignancy. CT imaging may be considered if there is suspicion of acute pulmonary embolism or an intraabdominal source for hemodynamic instability.

BLOODWORK AND URINE STUDIES

While neuroimaging studies evaluate structural pathology, bloodwork and urine studies screen for systemic disease processes, including infection and toxic-metabolic, and endocrine derangements. Bloodwork and urine studies are usually obtained in parallel to the history and clinical examination and often inform immediate management, as well as the direction for further workup.

EEG

The main applications for EEG in neurocritical care are as follows: (1) monitoring patients after prolonged convulsive status epilepticus to ensure adequate seizure control; (2) monitoring burst suppression in patients with refractory status epilepticus; (3) assessing for nonconvulsive seizures or nonconvulsive status epilepticus in patients with altered consciousness; and (4) clarifying the etiology of abnormal movements and spells.⁷⁰ EEG is also used as an adjunct tool in neurologic prognostication.⁷¹ Recent studies have shown correlations between EEG abnormalities and vasospasm and delayed cerebral ischemia in patients with subarachnoid hemorrhage.⁷² Additionally, the use of functional EEG in assessing

KEY POINTS

- Most critical structural intracranial pathologies are identified with a noncontrast head CT: hyperdensities, hypodensities, herniation, hydrocephalus, and skull fractures. CT angiography is often necessary to evaluate vascular pathologies. MRI of the brain can add valuable diagnostic and prognostic information but often does not change acute management.
- In the setting of trauma with impaired neurologic status, a CT of the cervical spine should be obtained to screen for fractures, misalignment of the bony spine, and evidence of cord compression.

patients with disorders of consciousness and detection of cognitive motor dissociation is being investigated.⁷³

In any patient with a decreased level of consciousness or fluctuating mental status, nonconvulsive seizures or status epilepticus should be excluded. Although the prevalence of seizures varies based on underlying pathology, the majority (up to 90%) of seizures in the intensive care unit are nonconvulsive, and EEG is necessary to detect them.⁷⁴ The suspicion for electrographic seizures is highest in patients with known acute or subacute cortical pathology and in those with extraaxial lesions (eg, subdural hemorrhage) adjacent to the cortex. A careful clinical examination can mitigate the concern for seizures and need for EEG; for example, a patient who is obtunded but able to track, follow commands, or communicate after vigorous stimulation is usually not in generalized status epilepticus.

In any patient with a high suspicion for seizures and a poor examination or any patient with unexplained altered mental status, EEG monitoring should be obtained. Continuous EEG monitoring is more sensitive in detecting seizures than a 30-minute routine EEG and, therefore, is generally the preferred study for critically ill patients.⁷⁵ Routine EEG may be sufficient in patients that exhibit frequent or continuous abnormal movements and can be captured immediately to clarify whether they have an electrographic correlate. Continuous EEG is more appropriate for patients with sporadic events, fluctuations over the course of the day, or persistently poor examination. With 24 hours of monitoring, the sensitivity to detect seizures in patients with acute brain injury is thought to be 95% for noncomatose patients and 80% for comatose patients, increasing to 98% and 87%, respectively, with 48 hours of monitoring.⁷⁴ The presence of interictal epileptiform discharges predicts an increased risk of seizure occurrence, warranting a longer EEG monitoring duration.⁷⁶ The duration of EEG monitoring should be determined based on the yield and evolution of the examination, and the need for imaging studies or procedures that may require interruption of the EEG recording, will have to be weighed based on treatment priorities.

LUMBAR PUNCTURE

In any patient with subacute or acute mental status changes or neurologic symptoms who also presents with fever, leukocytosis, or other clinical signs of infection, an LP should be strongly considered to assess for CNS infection (CASE 1-4). Other pertinent signs to determine the need for prompt LP and treatment include the presence of infectious prodrome (eg, headache, cough or congestion, gastrointestinal symptoms, another viral syndrome), overall immune status, exposure and travel, and other systemic signs of infection. Because CNS infections can be rapidly progressive, the threshold for LP should be low when there is clinical suspicion for infection or unexplained neurologic deterioration. LP should also be considered in patients with seizures of unclear etiology or unexplained confusion in the absence of clear infectious signs, especially if there is no clinical improvement and the initial workup is unrevealing.

In the presence of focal neurologic signs or a decreased level of consciousness, a head CT should be performed before the LP to gauge the risk of elevated ICP and potential herniation due to the procedure.⁷⁷ When LP cannot be immediately obtained because of concerns based on the CT or other logistical challenges related to the procedure, empiric treatment should be started

immediately. In the setting of bacterial meningitis specifically, delays in antibiotics are associated with increased mortality.⁴ Corticosteroids should also be considered before antibiotics in suspected bacterial meningitis.⁷⁸ Although the initiation of antibiotics may result in unrevealing cultures or changes in the CSF profile (higher glucose, lower protein), the overall pattern is still informative with a slightly delayed LP. Moreover, the white blood cell count and Gram stain are usually unchanged for the first 24 to 48 hours after steroids, atypical organisms may be detected, bacterial polymerase chain reaction and viral studies can be obtained, and the treatment plan can be adjusted based on positive results.

For patients with high clinical suspicion for intracranial aneurysmal rupture based on history (severe sudden-onset headache) or examination (third nerve palsy with an unreactive pupil, suggesting external compression by an adjacent posterior communicating artery aneurysm) but without subarachnoid blood on CT, LP is performed to assess for the presence of blood (consistently elevated red blood cell count between tube 1 and 4) and bilirubin (xanthochromia) in the CSF to confirm that a rupture has indeed occurred.

CSF studies are also part of the workup in suspected cases of Guillain-Barré syndrome. The typical finding of cytoalbuminologic dissociation as well as the presence of antiganglioside antibodies can support the diagnosis, but it is often not yet present in the early stages of the disease process.⁷⁹ Also, CSF studies should be considered when evaluating a suspected autoimmune, inflammatory, or paraneoplastic CNS process.

ELECTRODIAGNOSTIC STUDIES

Nerve conduction studies and EMG are adjunct diagnostic studies to inform diagnosis and localization in peripheral neuropathies, neuromuscular junction disorders, and muscle disease; for more information on these conditions, refer to the article “Neuromuscular Emergencies” by Catherine S. W. Albin, MD,⁴⁰ in this issue of *Continuum*. Electrodiagnostic results can confirm the underlying process and help the clinician assess whether confounding etiologies or alternative diagnoses may be present (eg, diabetic neuropathy, radiculopathy). Nerve conduction studies can also distinguish demyelinating from axonal neuropathies and polyradiculoneuropathies. Repetitive nerve stimulation and single-fiber EMG are important confirmatory tests for neuromuscular junction disorders. Often, depending on the availability of nerve conduction studies and EMG, treatment decisions must be based on the clinical picture while awaiting study results to further refine the plan.

CONCLUSION

Acute neurologic emergencies are time sensitive and require prompt examination, workup, and management. The workup is often fast paced with assessment and management frequently occurring in parallel. The evaluation begins with an assessment and stabilization of imminent threats to survival, including respiratory and hemodynamic compromise. A focused history and clinical examination build the initial differential diagnosis based on time course, constellation and evolution of symptoms, and localization of focal signs. Life-threatening, “must-not-miss” diagnoses should be rapidly considered and excluded.

KEY POINTS

- The majority of seizures in the neurocritical care unit are nonconvulsive. Continuous EEG should be considered in patients with an impaired or fluctuating level of consciousness based on the underlying diagnosis or when mental status changes are not fully explained.
- In patients with acute neurologic symptoms and clinical signs of infection, or those with unexplained altered mental status, LP should be considered. The LP should not delay empiric treatment for bacterial and viral central nervous system infection.

The clinical examination is often limited in critically ill patients with acute brain injury, but the priority is to identify findings that warrant emergent interventions, including signs of elevated ICP and herniation, acute neurovascular syndromes, clinical or subclinical seizures, and infections of the CNS. Clinicians must be mindful of the risk of clinical deterioration and facilitate appropriate monitoring and management. Scales and scores are helpful tools to assess disease severity, monitor evolution, and guide clinical decisions and prognostication, but they must be used in combination with a nuanced examination and interpreted with the overall clinical context in mind.

Neuroimaging should be obtained promptly in any suspected acute brain or spinal cord injury as part of the initial workup to identify critical structural abnormalities warranting urgent management. A comprehensive laboratory workup is needed to identify systemic (toxic-metabolic, infectious, and endocrine) processes. In any patient with an impaired level of consciousness, the EEG and LP should be considered.

Thoughtful and timely examination, workup, and management of patients in neurocritical care settings are imperative to optimize outcomes. Examination and workup should assess for diagnoses and complications warranting urgent management.

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